



Predictors in Adolescence of ESRD in Middle-Aged Men

Per-Ola Sundin, MD,^{1,2} Ruzan Udumyan, BSc, MSc,^{1,3} Per Sjöström, MD, PhD,¹ and Scott Montgomery, PhD^{1,3,4,5}

Background: Identification of predictors of end-stage renal disease (ESRD) in adolescence could provide intervention targets and improve understanding of the cause.

Study Design: Register-based nested case-control study.

Setting & Participants: A cohort of all Swedish male residents born from 1952 through 1956 who attended mandatory military conscription examinations in late adolescence was used to identify 534 cases and 5,127 controls matched by birth year, county, and vital status.

Predictor: Erythrocyte sedimentation rate (ESR), proteinuria, blood pressure, and body mass index (BMI) in late adolescence.

Outcomes: ESRD (defined here as dialysis therapy, kidney transplantation, surgical procedures creating long-term access for dialysis therapy, or chronic kidney disease stage 5) from 1985 through 2009.

Measurements: Physical working capacity and cognitive function score in late adolescence. Head of household's occupation and household crowding measured as person-per-room ratio from the 1960 census when participants were children.

Results: Proteinuria is associated notably with future ESRD, with an adjusted OR of 7.72 (95% CI, 3.94-15.14; $P < 0.001$) for trace or positive dipstick findings. ESR has a dose-dependent association with ESRD with an adjusted OR of 2.07 (95% CI, 1.14-3.75; $P = 0.02$) for ESR > 15 mm/h. Hypertension is associated strongly with future ESRD with an OR of 3.97 (95% CI, 2.08-7.59; $P < 0.001$) for grade 2 hypertension and higher. Elevated BMI is associated statistically significantly with increased ESRD risk with an OR of 3.53 (95% CI, 2.04-6.11; $P < 0.001$) for BMI ≥ 30 compared with 18.5- <25 kg/m².

Limitations: The study was limited to men, with no initial estimation of glomerular filtration rate, and information on smoking was unavailable.

Conclusions: ESR, proteinuria, BMI, and blood pressure in late adolescence are independent predictors of ESRD in middle-aged men. This highlights the long natural history and importance of adopting a life-course approach when considering the cause of chronic kidney disease.

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INDEX WORDS: End-stage renal disease (ESRD); erythrocyte sedimentation rate (ESR); proteinuria; body mass index (BMI); hypertension; adolescence; inflammation; disease trajectory; risk factor; etiology; kidney disease progression.

End-stage renal disease (ESRD) has significant effects for people with the disease in terms of life expectancy,¹ well-being, and difficulty continuing to work full-time.² The costs of treating chronic kidney disease (CKD) can be substantial, with some patients requiring long-term dialysis therapy and kidney transplantation.¹ The majority of individuals with ESRD have lost their kidney function in a gradual process, possibly over several decades. Identification of markers of the disease process in much earlier life not only adds to our understanding of the causes, but also may suggest intervention targets or identify individuals at increased risk for whom focused follow-up could be beneficial.

Diabetes³ and obesity⁴ are important risk factors for CKD, whereas hypertension^{5,6} and other cardiovascular diseases (CVDs)⁷ represent additional independent risks. Atherosclerosis leading to CVD and glomerulosclerosis resulting in CKD are to some extent parallel processes sharing risk factors, pathophysiologic mechanisms,^{8,9} and a role for inflammation.^{10,11} Atherosclerotic processes leading to CVD have their origins in adolescence and earlier life.¹²

Markers of risk prior to adulthood for subsequent CKD resulting in ESRD are less well described.

We examine markers in adolescence for their associations with ESRD in later adult life, up to 40 years later, in a longitudinal cohort of men in Sweden. The main markers in adolescence we used were hypertension; erythrocyte sedimentation rate (ESR) as a marker of inflammation, including from infections; proteinuria as a signal of early kidney damage; and body mass

From the ¹School of Health and Medical Sciences, Örebro University; ²Department of Medicine and ³Clinical Epidemiology and Biostatistics, Örebro University Hospital, Örebro; ⁴Clinical Epidemiology Unit, Karolinska University Hospital, Karolinska Institutet, Stockholm, Sweden; and ⁵Department of Epidemiology and Public Health, University College, London, United Kingdom.

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Address correspondence to Per-Ola Sundin, MD, Medicinska kliniken, Örebro University Hospital, 701 85 Örebro, Sweden. E-mail: perola.sundin@orebroll.se

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index (BMI). These were regarded as readily available markers of health (although another marker of inflammation would be used in a more modern setting) and development in adolescence, which theoretically could be associated with biological processes relevant to risk of future ESRD.

METHODS

Study Population

The population used for this study is a cohort comprising all men born between January 1, 1952, and December 31, 1956, who were included in the Swedish Military Conscription Register.^{13,14} At this time, assessment for compulsory military service was mandatory by law for all male Swedish citizens. Less than 4% of men with Swedish citizenship were excluded from the assessment that is undertaken at initial conscription due to chronic illness or disability. Examinations were conducted from 1970 through 1975, and the majority at ages 18 and 19 years, with a small number after this time at older ages.

The cohort consisted originally of 284,198 men, of whom 2,564 were excluded due to data inconsistencies such as incorrect personal identification number, female sex, or uncertain vital status. A further 225 men were excluded from the analysis due to improbable measurements at the conscription assessment (some exclusions were based on >1 anomaly): height < 144 cm (n = 39), BMI < 15 kg/m² (n = 134), weight > 178 kg (n = 9), systolic blood pressure < 50 or > 230 mm Hg (n = 33), and diastolic blood pressure < 30 or > 135 mm Hg (n = 12). Also excluded were those who did not complete the conscription examination due to chronic illness, disability, or lack of Swedish citizenship (n = 16,458).

Men with diagnoses at conscription consistent with existing kidney dysfunction or chronic inflammatory disease were excluded (n = 4,629): urogenital disease (n = 3,237); malformations of the kidneys and urinary tract (n = 140), diabetes (n = 554); other endocrine, metabolic, or nutritional disorders (n = 517); systemic inflammatory diseases (rheumatoid arthritis [n = 208], other collagenoses [n = 19], or vasculitis [n = 3]; see Table S1, available as online supplementary material, for diagnostic *International Classification of Diseases* codes). Exclusions for events prior to follow-up from January 1, 1985, for death (n = 2,963), emigration (n = 3,900), and ESRD diagnosis (n = 48) resulted in a sample of 253,430 men (there was some overlap in exclusions). We excluded 18,039 men with missing data for census (6,998) or any of the measures from the conscription assessment (12,261) that may have resulted from loss of records (some men were missing data in both categories). A total of 235,391 men were eligible as the sample for a nested case-control study.

Main Exposures

Data from the conscription examination in late adolescence were retrieved from the Swedish Military Conscription Register. Proteinuria was assessed using a simple dipstick test and classified as negative, trace, or positive. For those with proteinuria and no known associated conditions (an exclusion criterion), trace and positive were merged into a single category due to low numbers. ESR was determined by the Westergren method and was adjusted for erythrocyte volume fraction analyzed by the microhematocrit method consistent with the National Committee for Clinical Laboratory Standards. Systolic and diastolic blood pressure were measured after rest in a recumbent position using a sphygmomanometer and classified according to the World Health Organization as optimal (systolic < 120 and diastolic < 80 mm Hg), normal (120-129/80-84 mm Hg), high-normal (130-139/85-89 mm Hg), grade 1 hypertension (140-159/90-99 mm Hg), grade 2 hypertension (160-179/100-109 mm Hg), and grade 3 hypertension ($\geq 180/\geq 110$ mm Hg).¹⁵

Grades 2 and 3 hypertension were combined in a single category due to small numbers. BMI categories were calculated from weight and height measured in light clothing without shoes.

Covariates

The conscription examination also provided the following measures. Physical working capacity was assessed using an electronically braked ergometer by the maximum load sustained for 6 minutes, or an estimate derived from a nomogram for shorter durations. The original 0- to 9-point score was used to divide physical working capacity into 3 groups. A cognitive function assessment covering linguistic understanding, spatial recognition, general knowledge, and ability to follow mechanical instructions transformed into a single 9-point score was used and divided into 3 groups.

The 1960 census¹⁶ provided information for social and material circumstances during childhood. It provided data for head of household's occupation, characterized as manual/agricultural, farm owners/managers, office workers, business owners/managers, and others. A measure of household crowding was estimated as person-per-room ratio, divided into quarters of its distribution.

Outcome

The National Patient Register^{17,18} has recorded information for inpatient diagnoses and procedures since 1964, with the addition of outpatient records since 2001. This register was used to identify incident ESRD (defined here as long-term dialysis therapy, kidney transplantation, surgical procedures creating long-term access for dialysis, or a diagnosis of stage 5 CKD) from 1985 through 2009 (see Table S2 for the codes). To avoid results being driven by undiagnosed kidney disease in adolescence, we began follow-up in 1985, which is at least 10 years after the conscription examinations.

Statistical Methods

Information for the cases was used to identify suitable controls to conduct a nested case-control study. Controls were selected within the cohort with a ratio of 10 per case and matched for birth year, vital status, and county. Men with a cardiovascular diagnosis (Table S1) at the conscription assessment were excluded. Differences between characteristics of cases and controls were analyzed using *t* test for continuous variables and Pearson χ^2 test for categorical variables.

Conditional logistic regression was used to examine the association with ESRD for proteinuria, hypertension, ESR, and BMI. ESR was adjusted for erythrocyte volume fraction (continuous) in all analyses. In the adjusted model, all these measures were included together, with adjustment for physical working capacity, cognitive function score, and head of household's occupation and household crowding from the 1960 census.

Results are given as odds ratios (ORs) with 95% confidence intervals (CIs). *P* values were 2 sided and considered statistically significant at the 0.05 level. All analyses were conducted using Stata, version 12/SE for Windows (StataCorp LP).

RESULTS

From January 1, 1985, through December 31, 2009, a total of 534 incident ESRD diagnoses were identified and 5,127 men were selected as controls for creation of the nested case control study. Characteristics of cases and controls are listed in Table 1. Those with subsequent ESRD more often had a higher ESR and positive proteinuria test result and had higher blood pressure and BMI. They more often were from households with greater crowding in childhood, had lower cognitive function, and also had non-statistically significant

Table 1. Characteristics of Cases and Controls

	Cases (n = 534)	Controls (n = 5,127)	P
ESR category			0.07
1 mm/h	125 (23.4)	1,238 (24.1)	
2-6 mm/h	333 (62.4)	3,344 (65.2)	
7-10 mm/h	43 (8.1)	350 (6.8)	
11-14 mm/h	16 (3.0)	103 (2.0)	
≥15 mm/h	17 (3.2)	92 (1.8)	
Erythrocyte volume fraction (%)	46.7 ± 2.6	46.4 ± 2.4	0.009
Protein test result in urine			<0.001
Negative	517 (96.8)	5,104 (99.6)	
Positive	17 (3.2)	23 (0.4)	
Blood pressure category			<0.001
Optimal, <120/<80	78 (14.6)	987 (19.3)	
Normal, 120-129/80-84	153 (28.7)	1,735 (33.8)	
High-normal, 130-139/85-89	157 (29.4)	1,470 (28.7)	
Grade 1 HTN, 140-159/90-99	130 (24.3)	899 (17.5)	
Grades 2-3 HTN, ≥160/≥100	16 (3.0)	36 (0.7)	
BMI category			<0.001
Underweight, <18.5	50 (9.4)	623 (12.2)	
Normal weight, 18.5-25	387 (72.5)	4,148 (80.9)	
Overweight, 25-30	76 (14.2)	308 (6.0)	
Obese, ≥30	21 (3.9)	48 (0.9)	
Physical working capacity score			0.2
Lowest third	220 (41.2)	1,925 (37.5)	
Middle third	186 (34.8)	1,826 (35.6)	
Highest third	128 (24.0)	1,376 (26.8)	
Household crowding			0.005
Lowest quarter, 0.20-1.25	113 (21.2)	1,420 (27.7)	
Second quarter, 1.29-1.67	168 (31.5)	1,598 (31.2)	
Third quarter, 1.71-2.00	116 (21.7)	1,021 (19.9)	
Fourth quarter, 2.14-13	137 (25.7)	1,088 (21.2)	
Cognitive function score			0.001
Lowest third	139 (26.0)	1,062 (20.7)	
Middle third	285 (53.4)	2,684 (52.4)	
Highest third	110 (20.6)	1,381 (26.9)	
Head of household's occupation			0.006
Manual/agricultural worker	272 (50.9)	2,260 (44.1)	
Farm owner/manager	38 (7.1)	555 (10.8)	
Office worker	127 (23.8)	1,423 (27.8)	
Business owner/manager	60 (11.2)	555 (10.8)	
Other	37 (6.9)	334 (6.5)	

Note: Values for categorical variables are given as number (percentage); values for continuous variables, as mean ± standard deviation. Blood pressure (systolic/diastolic) is expressed as mm Hg; BMI, as kg/m²; household crowding, as person per room.

Abbreviations: BMI, body mass index; ESR, erythrocyte sedimentation rate; HTN, hypertension.

lower physical capacity in adolescence. Mean ages at conscription examination, start of follow-up, and end of follow-up were 19, 31, and 55 years, respectively.

In the unadjusted and adjusted models, proteinuria has a high-magnitude association with ESRD (Table 2), although the number of men with proteinuria at the conscription examination is small (Table 1). If trace and positive dipstick findings of proteinuria are analyzed separately, the adjusted ORs are very similar in magnitude (trace, 8.95 [95% CI, 2.89-27.7; $P < 0.001$]; positive, 7.16 [95% CI, 3.15-16.3; $P < 0.001$]). ESR is associated with ESRD in a dose-response manner, and this remains statistically significant after adjustment. Hypertension at conscription also is associated notably and statistically significantly with subsequent ESRD before and after adjustment. Higher BMI also is associated notably with ESRD (Table 2).

In the adjusted model, the 2 highest quarters of household crowding distribution and the lowest IQ score category were associated statistically significantly with increased risks for ESRD. Physical working capacity was not found to be associated statistically significantly with ESRD. Only having a head of household who was a farm owner/manager was associated statistically significantly with lower risk of ESRD compared with manual/agricultural work (adjusted OR, 0.60; 95% CI, 0.42-0.87; $P = 0.007$), with no other notable associations for parental socioeconomic index.

DISCUSSION

In this case-control study nested within a large and representative birth cohort of Swedish men, we found notable associations with ESRD in adulthood for measures collected in late adolescence of ESR, proteinuria, hypertension, and higher BMI. The associations could not be explained by contemporaneous morbidity in adolescence. We excluded ESRD diagnoses in the 10- to 15-year period between examination in adolescence and the beginning of follow-up, further emphasizing the long natural history of ESRD.

Several inflammatory markers, including C-reactive protein and fibrinogen, have been associated with risk of incident CVD.^{19,20} In a recent study of 433,577 Swedish conscripts by Toss et al,²¹ ESR was associated with myocardial infarction up to 40 years later. Because CVD and CKD share several risk factors and are risks for each other, similar long-term associations are likely for CKD.^{7,22,23} After CKD onset, the prevalence of elevated levels of inflammatory markers is high.²⁴ The processes of atherosclerosis and glomerulosclerosis have pathophysiologic similarities,^{8,9} and inflammation is an important factor in both processes.^{10,11} Thus, early identification of

Table 2. Associations of Clinical and Demographic Characteristics in Adolescence With ESRD in Middle-Aged Men

	Unadjusted		Adjusted ^a	
	OR (95% CI)	P	OR (95% CI)	P
ESR category				
1 mm/h	1.00 (reference)		1.00 (reference)	
2-6 mm/h	1.13 (0.90-1.43)	0.3	1.10 (0.87-1.40)	0.4
7-10 mm/h	1.53 (1.04-2.26)	0.03	1.39 (0.94-2.08)	0.1
11-14 mm/h	2.03 (1.14-3.63)	0.02	1.93 (1.06-3.50)	0.03
≥15 mm/h	2.46 (1.39-4.36)	0.002	2.07 (1.14-3.75)	0.02
Proteinuria				
Negative	1.00 (reference)		1.00 (reference)	
Positive	7.70 (4.02-14.75)	<0.001	7.72 (3.94-15.14)	<0.001
Blood pressure category				
Optimal, <120/<80	0.90 (0.67-1.19)	0.4	0.95 (0.71-1.26)	0.7
Normal, 120-129/80-84	1.00 (reference)		1.00 (reference)	
High-normal, 130-139/85-89	1.21 (0.96-1.53)	0.1	1.19 (0.93-1.51)	0.2
Grade 1 HTN, 140-159/90-99	1.66 (1.29-2.13)	<0.001	1.45 (1.12-1.88)	0.005
Grades 2-3 HTN, ≥160/≥100	5.16 (2.79-9.56)	<0.001	3.97 (2.08-7.59)	<0.001
BMI category				
Underweight, <18.5	0.86 (0.64-1.17)	0.4	0.83 (0.61-1.15)	0.3
Normal weight, 18.5-<25	1.00 (reference)		1.00 (reference)	
Overweight, 25-<30	2.63 (2.00-3.44)	<0.001	2.36 (1.79-3.13)	<0.001
Obese, ≥30	4.73 (2.81-7.96)	<0.001	3.53 (2.04-6.11)	<0.001
Physical working capacity				
Lowest third	1.23 (0.98-1.56)	0.07	1.23 (0.96-1.57)	0.1
Middle third	1.09 (0.86-1.39)	0.5	1.08 (0.85-1.38)	0.5
Highest third	1.00 (reference)		1.00 (reference)	
Household crowding				
Lowest quarter, 0.20-1.25	1.00 (reference)		1.00 (reference)	
Second quarter, 1.29-1.67	1.33 (1.04-1.71)	0.02	1.28 (0.98-1.65)	0.07
Third quarter, 1.71-2.00	1.45 (1.10-1.90)	0.008	1.34 (1.01-1.79)	0.04
Highest quarter, 2.14-13.00	1.60 (1.23-2.09)	<0.001	1.46 (1.10-1.94)	0.009
Cognitive function score				
Lowest third	1.66 (1.28-2.17)	<0.001	1.33 (1.00-1.77)	<0.05
Middle third	1.35 (1.07-1.70)	0.01	1.20 (0.94-1.52)	0.1
Highest third	1.00 (reference)		1.00 (reference)	

Note: Blood pressure (systolic/diastolic) is expressed as mm Hg; BMI, as kg/m²; and household crowding, as person per room.

Abbreviations: BMI, body mass index; CI, confidence interval; ESR, erythrocyte sedimentation rate; ESRD, end-stage renal disease; HTN, hypertension; OR, odds ratio.

^aAdjusted for socioeconomic factors from the 1960 census (head of household's occupation, household crowding) and characteristics in adolescence (ESR, erythrocyte volume fraction, proteinuria, blood pressure, BMI, physical working capacity, and cognitive function score).

increased levels of inflammatory markers may be useful in predicting risk of future CKD.

Several studies have examined the association of inflammatory markers with CKD, but these have tended to be samples defined by CVD or CKD, rather than a well-characterized general population sample followed up longitudinally, as studied here. In the ARIC (Atherosclerosis Risk in Communities) Study, higher values for white blood cell count, von Willebrand factor, factor VII, and fibrinogen were associated with lower estimated glomerular filtration rates (eGFRs).²⁵ Participants in the ARIC cohort were aged 45-64 years at baseline and mean follow-up was

14 years. Another population-based study from Canada found that higher values for tumor necrosis factor α receptor 2, interleukin 6 (IL-6), and white blood cell count were associated with lower eGFRs.²⁶ These 43- to 84-year-olds were followed up for 15 years. This study did not have information for proteinuria. Participants aged 45-84 years in MESA (Multi-Ethnic Study of Atherosclerosis) without CKD or CVD were followed up for 5 years, and associations were observed for IL-6, factor VIII, and C-reactive protein levels with rapid decreases in kidney function (eGFR, -3 mL/min/1.73 m² per year).²⁷ Only IL-6 level was associated significantly with low eGFR.

Our results for ESR as a marker of inflammation are consistent with earlier studies linking inflammation with ESRD risk. In contrast to earlier studies, the initial examination was in late adolescence and with a delayed start of follow-up: this demonstrates the presence of low-grade inflammation at the time of transition from childhood to adult life in those who will develop ESRD up to 40 years later. No exclusion was made at the conscription examination for transient infections or other inflammatory conditions, which suggests that the association with chronic low-grade inflammation might be underestimated.

Proteinuria signals glomerular vascular damage and also is a potent risk factor for CVD.²⁸ Recent meta-analyses from the CKD Prognosis Consortium using pooled data from 26 adult general population cohorts including more than 1.8 million participants show a strong association between proteinuria and ESRD across a wide range of ages²⁹ and with no evidence of a sex difference.³⁰ Our results are consistent with earlier studies and further emphasize the long natural history of ESRD. This suggests the importance of follow-up of younger people with unexplained proteinuria because men with relevant diagnoses in adolescence were excluded in our study. The use of proteinuria to screen for future CKD has not been evaluated in randomized controlled trials.³¹

The risks associated with trace and positive dipstick findings in our study were of approximately the same magnitude, but given the low numbers in each group and the possibility of low sensitivity in the dipstick test, it is not reasonable to make further conclusions on which proteinuria level constitutes a significant risk for ESRD in this setting.

Adult hypertension is an established major risk factor for CKD.^{5,6} In a recent large meta-analysis, it was concluded that even high-normal blood pressure may indicate an increased risk of future ESRD compared with optimal blood pressure.³² To our knowledge, the association of primary hypertension in adolescence with future CKD has not been studied. We found an elevated risk for ESRD beginning at higher normal blood pressures and reaching statistical significance for grade 1 hypertension.

Several studies have demonstrated that elevated BMI is a risk factor for future CKD in adults,⁴ and childhood obesity also has been linked to adult CKD.³³ This association has been studied in a setting similar to our study examining 1.2 million Israeli conscripts.³⁴ In that study, the association was significant for all-cause ESRD and ESRD with and without diabetes, with considerably higher hazard ratios for ESRD with diabetes. Higher BMI is likely to influence ESRD risk through cardiovascular mechanisms and type 2 diabetes.

Although to our knowledge, no large long-term prospective study has evaluated the role of socioeconomic

factors in predicting CKD and ESRD, some retrospective and cross-sectional studies have linked socioeconomic factors including education, income, and socioeconomic indexes with CKD.³⁵⁻³⁸ We included head of household's occupation and household crowding in our models to account for socioeconomic conditions that may be potential confounding factors. For this reason, we also included cognitive function score in our adjusted model because it is related to educational attainment and many socially mediated risks.

Physical inactivity has been associated with albuminuria³⁹ and CKD^{40,41} in some studies, but the results are not consistent. In a recent large Korean study by Park et al,⁴² low physical working capacity was associated with glomerular hyperfiltration and albuminuria, which are early markers of CKD. The findings were restricted to male participants and the association with hyperfiltration was significant for only the younger participants. A physical working capacity measure was included in our models for adjustment for personal characteristics; in nominal terms, lower capacity scores tended to associate with increased risk for ESRD, but this effect was not statistically significant.

Our study has several potential important limitations. Because the baseline examination was based on conscription for military service, only men are included because conscription was limited almost exclusively to men at this time. There is no evidence of a sex difference in the relationship between proteinuria and ESRD,³⁰ but inclusion of only men limits the generalizability of our results. The initial assessment at conscription did not include a measure of GFR, which means there may have been undiagnosed CKD at baseline. However, established CKD is rare in late adolescence, and we excluded those with potentially relevant diagnoses and delayed start of follow-up 10-15 years after the initial assessment. It is notable that ascertainment of ESRD should be largely complete because it almost always results in hospital admission, and because we have records from all Swedish hospitals, it is unlikely that we could have missed a meaningful number of cases. Although hypertension in adolescence is associated with increased ESRD risk, we do not know if treatment of hypertension was initiated in adolescence, to what extent that would ameliorate this risk, or how many of the men with ESRD developed hypertension in adulthood. Another limitation is the lack of detailed information for the properties of the dipsticks used for proteinuria assessment and the timing of the sample because this information is regarded as classified by the Swedish military. We could not take smoking into account and this is an important risk factor, but one unlikely to account entirely for our reported associations. The exclusion of participants in the cohort

because of incomplete data potentially could have introduced selection bias.

Our results support the concept that the origin of CKD often can be traced back into childhood much in parallel to the evidence supporting the onset of atherosclerosis early in life.¹² Apart from prematurity, low birth weight, and rapid weight gain in childhood, it is largely unknown which early life circumstances imply an elevated risk of CKD.⁴³ Given the associations with ESR, proteinuria, BMI, hypertension, and other markers in late adolescence, this further suggests that the origins of ESRD may be earlier in childhood.

In summary, we have found notable associations for ESR, BMI, proteinuria, and hypertension in late adolescence with ESRD in middle-aged men up to 40 years after initial assessment. This evidence of a long natural history highlights the value of adopting a life-course approach to understanding and preventing CKD.

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Contributions: Research idea and study design: P-OS, SM; data acquisition: SM; data analysis/interpretation: P-OS, SM, RU, PS; statistical analysis: P-OS, RU; supervision or mentorship: SM, PS. Each author contributed important intellectual content during manuscript drafting or revision and accepts accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any portion of the work are appropriately investigated and resolved. P-OS takes responsibility that this study has been reported honestly, accurately, and transparently; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned have been explained.

SUPPLEMENTARY MATERIAL

Table S1: Diagnostic codes at conscription examination.

Table S2: Definition of ESRD.

Note: The supplementary material accompanying this article (<http://dx.doi.org/10.1053/j.ajkd.2014.06.019>) is available at www.ajkd.org

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